



Voting System Anomaly Root Cause Analysis Template v2.0

Root Cause Analysis for:

**VV40ECT-242: VxMarkScan ESD Secure Boot Violation Message Required Human
Intervention to Continue
VxSuite, Version 4.0 and EAC Certification #VXS4**

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Introduction

This RCA documents investigation and mitigation of an issue arising during standard electrostatic discharge (ESD) testing of VxMarkScan with SLI Compliance

Anomaly Description

Complete all sections. Descriptions must be as detailed as possible, while being clear and concise since the anomaly is the source of the entire RCA. This detail should include a complete list and/or description of the “symptoms” of the anomaly and the conditions present which the symptoms occurred.

<u>Date of Anomaly:</u> April 14, 2026	<u>Time of Anomaly:</u> 1:31pm
<u>Place of Anomaly:</u> Element, Longmont, CO	<u>Person identifying Anomaly:</u> Jessica Myers, VotingWorks
<u>Expected Results of actions leading up to anomaly:</u> Any system errors in VxMarkScan that result in showing a “Something went wrong” error message in the official software should lead to an automatic reboot. The reboot should occur without issue, reloading the VxMarkScan software as normal without human intervention. This includes when responding to any major ESD onto the system that might require a reboot.	
<u>Detailed description of the event / anomaly:</u> VxMarkScan unit MK-11-004 was undergoing standard ESD testing, where contact discharges were applied on the exposed strip of metal on the PAT jack plugged into the PAT port (prior to mitigations). When applying that ESD, the system detected resulting internal errors that triggered a “Something went wrong” message to display on the screen, followed by an attempt to automatically reboot. This automatic reboot was the intended solution to such an error. However, instead of booting normally without human intervention, the screen then showed a red dialog with the headline “Secure Boot Violation” and message “Invalid signature detected. Check Secure Boot Policy in Setup,” indicating that an incorrect software image was attempting to boot. Continuing through the dialog manually and pressing “OK” allowed the user to choose the correct software image to boot that meets Secure Boot requirements.	

EAC (Election Assistance Commission) Root Cause Analysis

If the anomaly is repeatable, provide step by step instructions to recreate it:

1. Set up the VxMarkScan unit for testing running the Hardware Test Utility software.
2. To reproduce the issue, the unit should also have multiple drives or images installed on the system. Under administrator privileges in the BIOS, alter the boot order to ensure the incorrect image boots before the official software.
3. Conduct standard ESD testing, applying -8kV contact ESD to the interface of PAT jack inserted into the PAT port. Observe software behavior. If the error message "Something went wrong" appears on screen, observe the system attempt to automatically reboot. Upon rebooting, a "Secure Boot Violation" error dialog will show.

Chronology of Events / Timeline

Provide a detailed chronology of the events leading up to, and following, the anomaly. Add additional events if necessary.

ID	Date/Time	Description	Entity Org/person	Result / Notes
1	4/14/26, 1:31pm Mountain Time	VxMarkScan undergoes ESD testing. An anomaly is reported involving the automatic restart feature not functioning properly.	SLI, and Jessica Myers, VotingWorks	VxMarkScan unit MK-11-004 was undergoing ESD testing, when -8kV contact discharge on the interface between the PAT jack and PAT port caused a "Something went wrong" error that required an automatic reboot. Upon starting the reboot, the VxMarkScan showed a "Secure Boot Violation" message. VotingWorks analyzed more details about the unit, its test history, and discussed the potential causes. Install logs were also collected for analysis.
2	4/14/26 3:20pm	VotingWorks analyzed install logs to confirm multiple drives were on the unit.	Adam McManus, Arsalan Sufi, VotingWorks	The behavior was theorized to come from known causes where the system has multiple boot entries and an incorrect boot order and tries to access the wrong volume. Logs also showed that two conflicting Logical Volume Managers were present. It was noted that no other USB was inserted in the system during testing, like in elections. Therefore another system was installed internally. Possible fixes were discussed, including wiping the internal eMMC and reimaging the machine with the proper test software. Plans were made to fix the system software at a later date, due to limited time with the test lab.

EAC (Election Assistance Commission) Root Cause Analysis

3	4/22/26 1:58pm	The VxMarkScan unit boot entries and boot order were corrected.	SLI, with Arsalan Sufi, VotingWorks	The boot entries were analyzed in the system. The default Debian entry in the unit was removed from the boot order, ensuring the VotingWorks application was booted first. In addition, the system reimaged more times. This confirmed that subsequent imaging operations kept the VotingWorks application as the first in the boot order, rather than reverting to the default Debian entry.
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Investigative Team and Method

This section shall describe how the investigative team is assembled by the voting system manufacturer, who it consists of, and how it gathers the data to be used in the analysis. Include the RCA method employed by the manufacturer in conducting the analysis and why this method was used.

Names and Positions of members of the investigation team:
Jessica Myers, Head of Compliance
Adam McManus, Production Engineer
Arsalan Sufi, Head of Software
Describe the data gathering process:
Jessica Myers and initiated the investigation as the VotingWorks staff on-site at Element during these tests. They received reports from SLI/Element about the anomaly and started initial troubleshooting while coordinating the response. Arsalan Sufi and Adam McManus investigated the issue further by analyzing logs and working with SLI staff to properly reimage the unit.

EAC (Election Assistance Commission) Root Cause Analysis

Describe which methodology(s) is used to conduct the root cause analysis:

A limited “5 Why’s” approach was used for this investigation, quickly leading to a focus on the boot order due to it being a known issue. The series of questions were:

- Why would the Secure Boot Violation error show up, and inconsistently?
- Why would the wrong boot entry load, instead of the proper image?
- Why would the boot entries be out of order?

This approach informed the actions to analyze logs and test histories for the unit, leading to the findings here.

Findings and Root Cause

Describe the findings of the investigation and explain the root cause(s) based on these findings. If the RCA results in findings that are not directly related to the root cause of the anomaly, these should also be captured as manufacturer product/process improvement steps in an effort to improve the voting system.

The BIOS on VxMarkScan systems ensures at least one default entry for a Debian system is present. During earlier testing and imaging on the MK-11-004 unit, that default entry was inserted in the boot order ahead of the installed VotingWorks systems. Since that default entry does not meet VotingWorks Secure Boot requirements, the red warning dialog is displayed. When a user manually continues through that warning dialog, the secure VotingWorks system is then successfully booted.

To correct the issue, the default Debian entry was manually removed from the boot order, ensuring the VotingWorks application was booted first. In addition, it was confirmed that subsequent imaging operations kept the VotingWorks application as the first in the boot order, rather than reverting to the default Debian entry.

Finally, correcting these boot entries enables the expected response to major ESD by triggering automatic reboots without manual intervention.

Corrective Action(s)

The process of corrective actions started by accessing the BIOS and boot entries of the VxMarkScan test unit using secured administrator privileges. The boot entries in the unit were corrected, by removing the default Debian boot entry and giving the proper VotingWorks image boot priority. VotingWorks staff walked through this corrective actions process with SLI via phone call, while SLI completed the actions on the unit directly.

No other corrective actions are required, since installation tools for VotingWorks software will define the correct boot order without manual intervention. Changing this boot order requires administrator privileges in the BIOS.

Solution Management

The purpose of this section is to manage the corrective action(s) moving forward. This should detail all process changes to manage those corrective actions, and steps taken to ensure the actions eliminate the anomaly over time.

Since boot entry order is defined by administrators in the BIOS, this will never be changed except by those with BIOS access and during certification testing. Instruction can be provided as needed in certification test campaigns on how to correct the boot order or boot entries if they have been changed during testing.